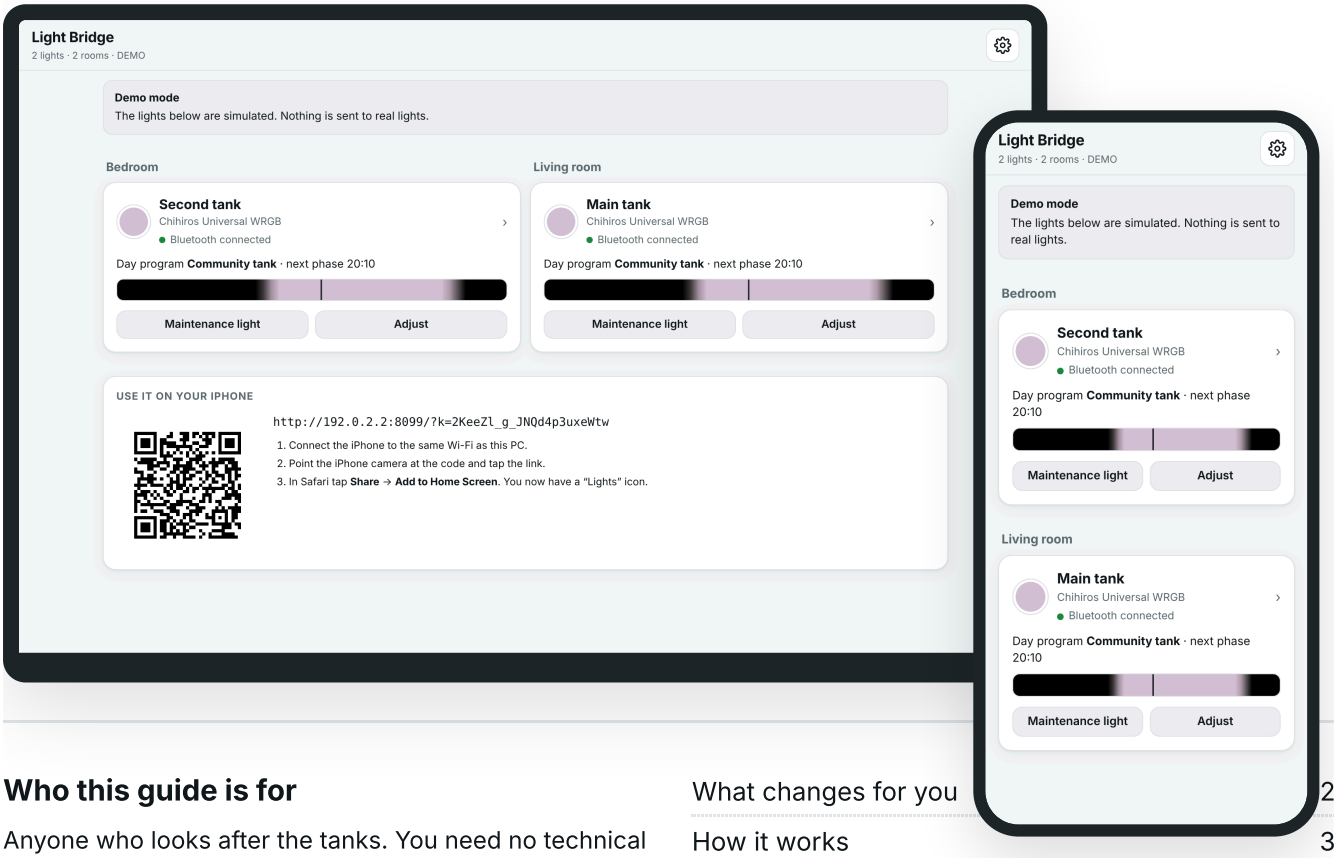
GUIDE · VERSION 0.11 · OCTOBER 2026

Light Bridge

Your Chihiros lights, made simple

Set the day light of your **Chihiros Universal WRGB** lights phase by phase, on the PC and on your iPhone. The program is stored in each light, so the lights keep running when the laptop goes to work.



Who this guide is for

Anyone who looks after the tanks. You need no technical knowledge. Steps marked **ONE TIME** are done once and never again.

The pictures show the demo mode with simulated lights. Your screen looks the same, with your own lights and rooms.

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OVERVIEW

What changes for you

Light Bridge is a small program for your laptop. It gives you one clear page for both Chihiros lights and stores your day programs in the lights themselves.

BEFORE (MY CHIHIROS APP)	WITH LIGHT BRIDGE
The colour steps over the day can only be set all at once.	→ Every phase of the day is changed on its own. The tank shows the result right away.
The app often loses the connection.	→ The lights run their program on their own. The laptop is only needed for changes.
Two lights in two rooms, set up one by one.	→ One page for both, sorted by room. Each light keeps its own program.
After a power cut, the light can have the wrong time.	→ Light Bridge sets the clock of each light whenever the laptop is near it.
Manual colours stay on until someone switches back.	→ Manual light ends on its own after the time you choose.
Account, cloud, app updates.	→ No account, no cloud. Everything stays in your home.

Easy to use

The menu follows the Juwel HeliaLux, which many people find clear and simple:

Phases

A day is a list of phases. Each phase says: “at this time, reach these colours”. In between, the light fades.

Live preview

While you change a phase, the tank shows it immediately. A slider lets you look at any time of day.

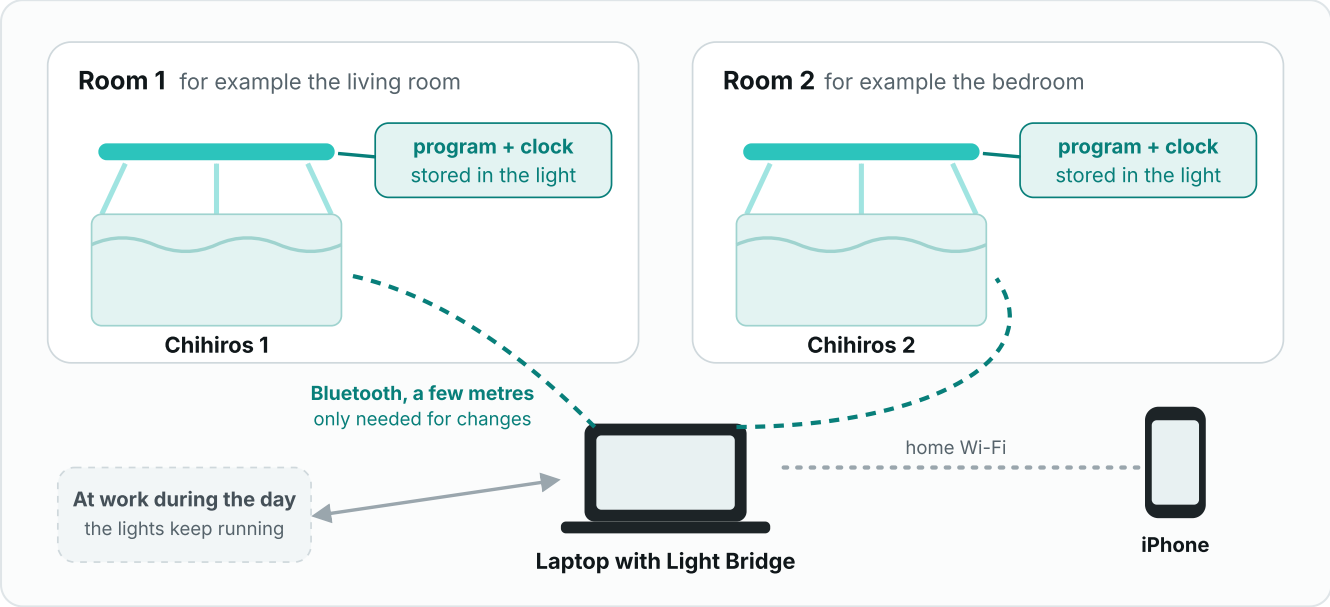
Templates

Ready-made day programs, for example with a midday break or a warm sunset. Pick one and adjust it.

THE IDEA

How it works

The day program lives **in the light**. The laptop is like a remote control: you only need it nearby to change something.



Three simple rules

1. **Each light runs on its own.** It keeps its day program and its clock. It needs neither the laptop nor the Wi-Fi to run.
2. **The laptop is only needed for changes.** To store a change, set the clock or switch on manual light, the laptop must be within a few metres of that light. Take it to work as usual.
3. **The iPhone works at home.** It talks to the laptop through your home Wi-Fi. The laptop must be at home, switched on and running Light Bridge.

When the laptop leaves

The lights simply keep running their stored programs. Only a **manual light** that is on at that moment stays on until the laptop is back. Light Bridge shows this clearly (see page 7). Tip: tap **Back to day program** before you leave.

SETUP ONE TIME

Setting up the PC

A Windows laptop	With Bluetooth. Almost every laptop has it. It is used as a normal laptop; it may go to work, sleep or be switched off.
Your two Chihiros lights	Switched on, as usual.
The Light Bridge folder	The file `light-bridge.zip` you received with this guide.
About 20 minutes	Only once.

1. **Unpack the folder**
   Right-click `light-bridge.zip` → **Extract All...** and choose your **Documents** folder. You now have a folder called **light-bridge**.

2. **Install Python (free)**
   Light Bridge needs a free helper program called Python. Go to `python.org/downloads` and click the big yellow **Download Python** button. Open the downloaded file. **Important:** at the bottom of the first window, tick **Add python.exe to PATH**, then click **Install Now**.

3. **Start Light Bridge**
   Open the **light-bridge** folder and double-click `Start-Light-Bridge`. If Windows says "Windows protected your PC", click **More info** → **Run anyway**. The first start takes about one minute.

4. **Allow network access**
   Windows asks whether Python may use the network. Tick **Private networks** only and click **Allow**. This lets your iPhone reach the laptop at home.

5. **The page opens**
   Your browser opens Light Bridge. Continue with "Adding your lights" on the next page.

Recommended: start with Windows
Double-click `Start-With-Windows` once. Light Bridge then starts by itself, minimised, when you sign in, and is ready whenever the laptop is at home. To undo: `Stop-Starting-With-Windows`.

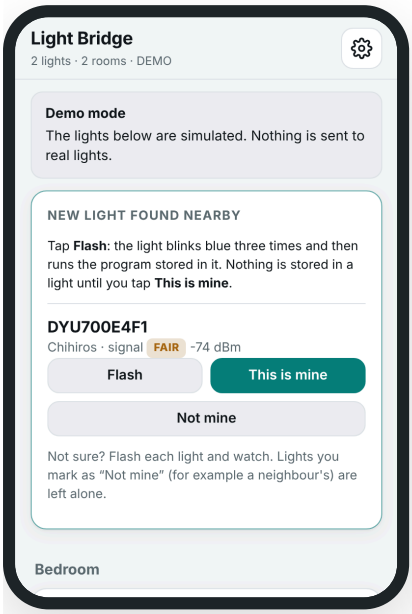
The black window
Light Bridge runs in a black window. You can minimise it. If you close it, Light Bridge stops; the lights keep running their stored programs.

Sleep is fine. You do not need to keep the laptop awake: the lights do not need it.

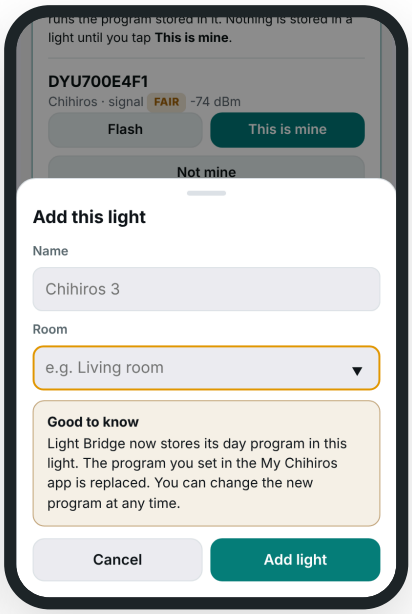
SETUP ONE TIME PER LIGHT

Adding your lights

Light Bridge finds Chihiros lights nearby by itself. To be safe, it changes nothing in a light until you confirm that it is yours.



A new light nearby



Name and room

1. **Go to the first light**
   Take the laptop into the room, within a few metres of the light.
2. **Wait for the card**
   Within a minute, "New light found nearby" appears.
3. **Tap Flash**
   The light blinks blue three times. Is it the one in front of you?
4. **Tap This is mine**
   Enter a name and the room, then **Add light**.
5. **Second light**
   Walk to the other room and do the same.

Good to know

When you add a light, Light Bridge stores its own day program in it (it starts with the template "Community tank"). The program you had set in the My Chihiros app is replaced. You can change the new one at any time.

Not your light?

Tap **Not mine**, for example for a neighbour's Chihiros. Light Bridge then leaves that light alone and does not show it again.

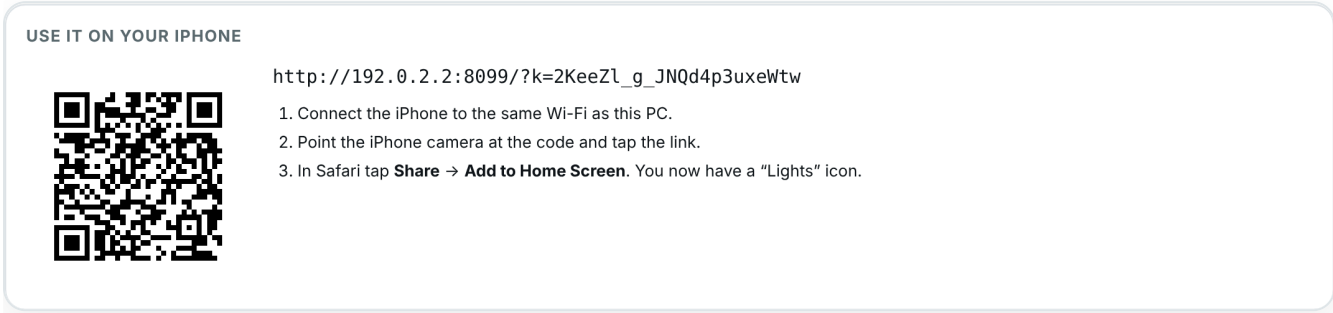
Close the old app

A Chihiros talks to only one device at a time. Close the My Chihiros app on every phone in the household, otherwise it may block Light Bridge.

SETUP ONE TIME

Using it on the iPhone

No App Store download is needed. The iPhone opens the page that the laptop offers in your home Wi-Fi.



At the bottom of the overview on the laptop: the code for your iPhone.

1 **Same Wi-Fi**

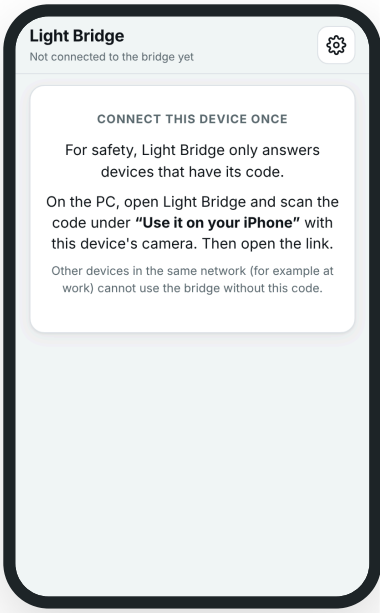
The iPhone must be in your home Wi-Fi (not on mobile data only), and the laptop at home with Light Bridge running.

2 **Scan the code**

Open the iPhone camera and point it at the code on the laptop screen. Tap the link that appears. Safari opens Light Bridge.

3 **Add it to the Home Screen**

In Safari, tap **Share** → **Add to Home Screen** → **Add**. You now have an icon called **Lights**, like an app.



Without the code

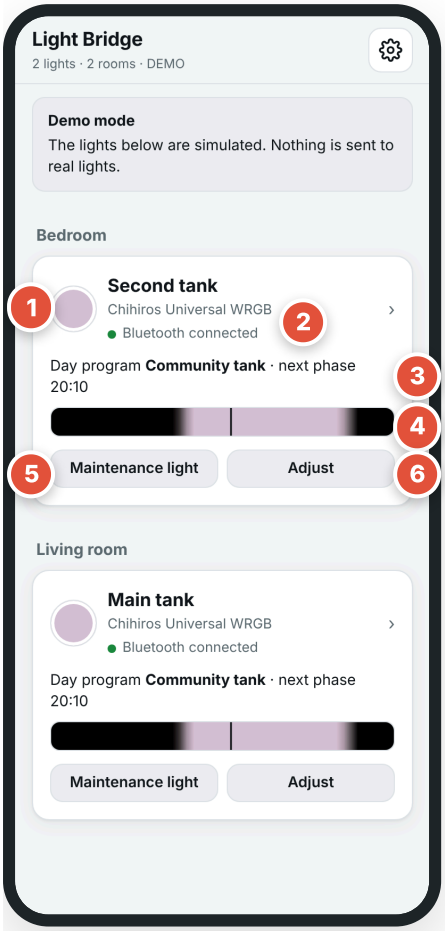
Why a code?

The code contains a secret key. Only devices that scanned it can use Light Bridge. Other people in the same network, for example when the laptop is at work, cannot. If a phone shows “Connect this device once”, simply scan the code again.

EVERYDAY USE

The overview

The first page shows all lights, sorted by room. One look tells you what each light is doing.



1. **Colour dot.** What the light shows right now.
2. **Connection.** See the table below.
3. **What is running.** The day program and the next change, or for example “Manual light until 14:30”.
4. **Today's colour bar.** What the light runs today, from 0:00 to 24:00. The thin line is the current time.
5. **Maintenance light.** Full light for 60 minutes, for cleaning or planting. It ends on its own.
6. **Adjust.** Change colours now, edit the day program, or change the settings.

● **Bluetooth connected**	The laptop is near. Changes go through right away.
● **Out of range · runs its own program**	All good. The laptop is away; the light runs its program.
● **Out of range · still in manual light**	The manual light stays until the laptop is near this light again.
CHANGE WAITING	A change is stored as soon as the laptop is near the light.

EVERYDAY USE

Changing the light right now

Tap a light. The tab **Now** shows what the light is doing and lets you change it at once.

Manual light

Move a slider. The light changes immediately. Each slider is one colour: **W** white, **R** red, **G** green, **B** blue. 0 is off, 100 is full.

Below the sliders, choose how long: 30 minutes, 1 hour, 3 hours or 12 hours. After that, the light goes back to its day program on its own.

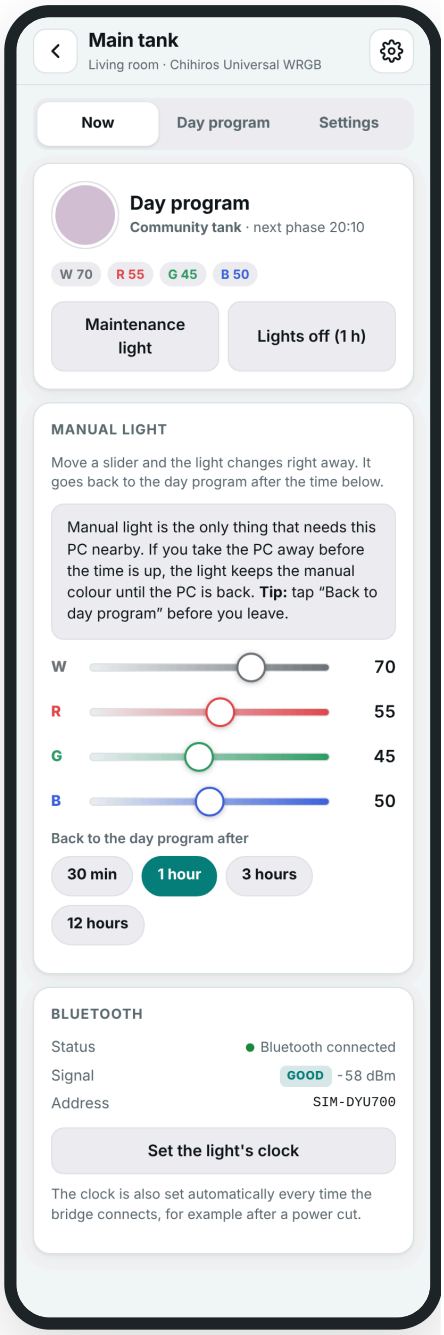
Quick buttons

Maintenance light	All colours at full power for 60 minutes.
Lights off (1 h)	Dark for one hour, then back to the day program.
Back to day program	Ends manual or maintenance light at once.

Manual light needs the laptop nearby to end

The light can end its day program on its own, but not a manual light. Only the laptop can switch it back. If you take the laptop away before the time is up, the manual light stays on until the laptop is near the light again.

Before you leave: tap **Back to day program**.

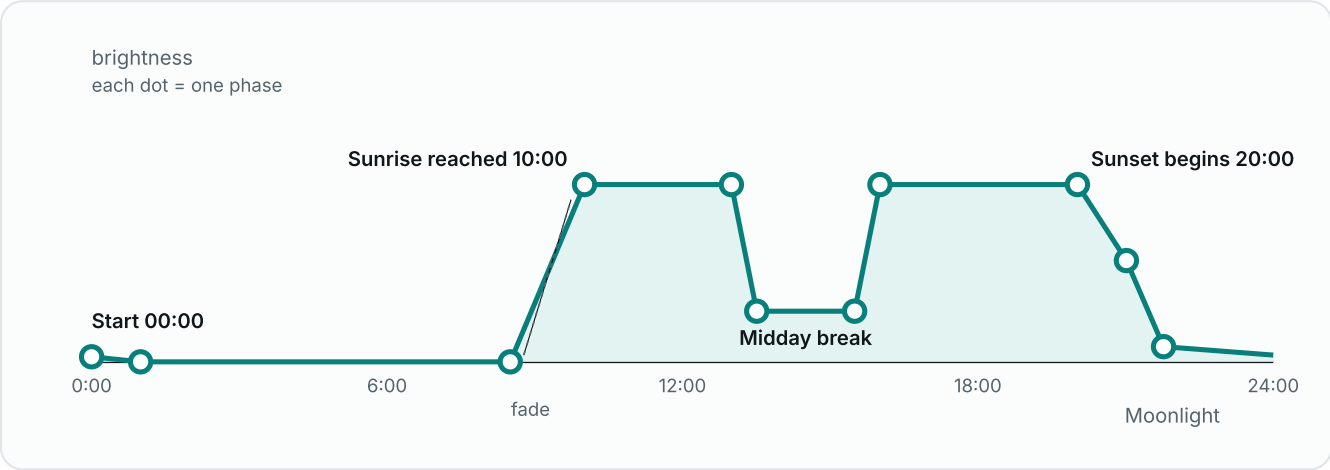


Tab "Now"

DAY PROGRAM

How a day program works

A day program is a list of **phases**. Each phase says: “at this time, the light should have reached these colours”. Between two phases, the light fades smoothly.



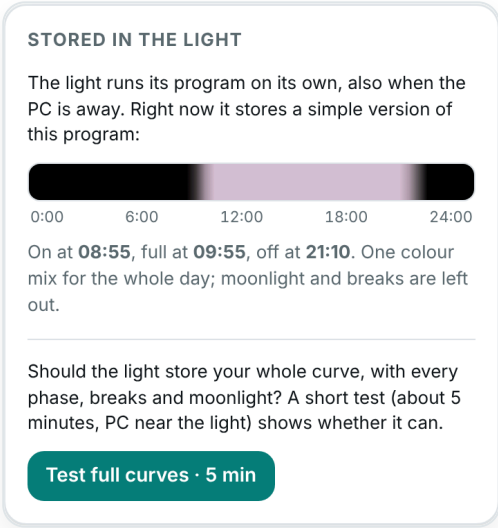
The rules

- The first phase is always at **00:00**, so the night is always defined.
- To hold a level, give two phases the same colours (10:00 and 13:00 above).
- Each light keeps **8 day programs**. You choose which runs on which weekday.

What the light stores: the simple version

The Chihiros can store a simple program by itself: **switch on, fade up, full light, fade down, off**, with one colour mix. Light Bridge makes this simple version from your day program and stores it in the light. Moonlight and a midday break are left out. The card “Stored in the light” shows exactly what the light runs.

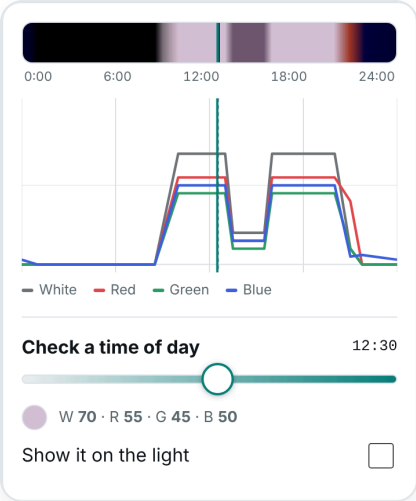
If the light can also store **full curves** with all phases, the light runs your whole program. A short test shows whether it can (page 11).



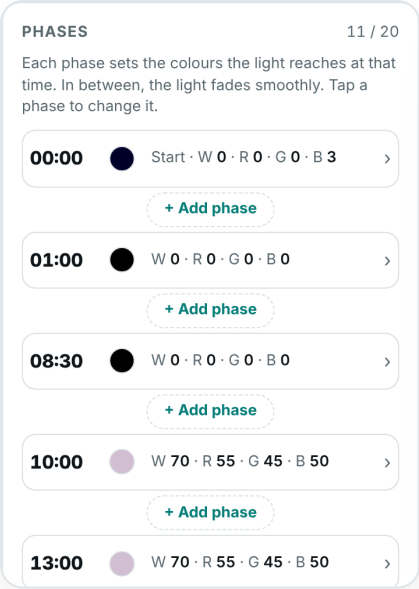
The card “Stored in the light”

DAY PROGRAM

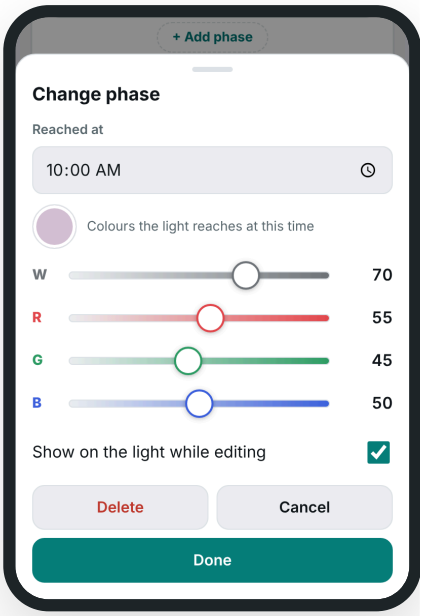
Changing the day program



Colour bar, chart and “Check a time of day”



The phases, with “+ Add phase”



Changing one phase

1 Open the day program

Tap the light → tab **Day program**. At the top, choose which of the 8 programs you want to change. “(runs today)” marks today's program.

2 Change a phase

Tap a phase. Set the time and the colours. With **Show on the light while editing** on, the tank shows the colours right away. Tap **Done**.

3 Add or remove phases

+ Add phase between two phases adds one in the middle. **Delete** in a phase removes it.

4 Check the whole day

Move the slider **Check a time of day**. Tick **Show it on the light** to see that moment in the tank.

5 Save

Tap **Save to light** in the bar at the bottom. If the laptop is near the light, the program is stored after a few seconds. If not, it shows CHANGE WAITING and is stored the next time the laptop is near.

Templates and copying

Start from a template loads a ready-made day. **Copy to another light** puts the same day on the other Chihiros.

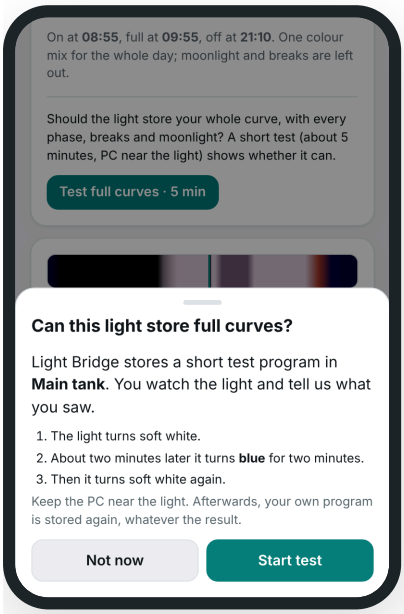
Weekdays

Under “Used on”, tap the days this program should run, for example a later sunrise at the weekend. (With full curves, every day uses the same program.)

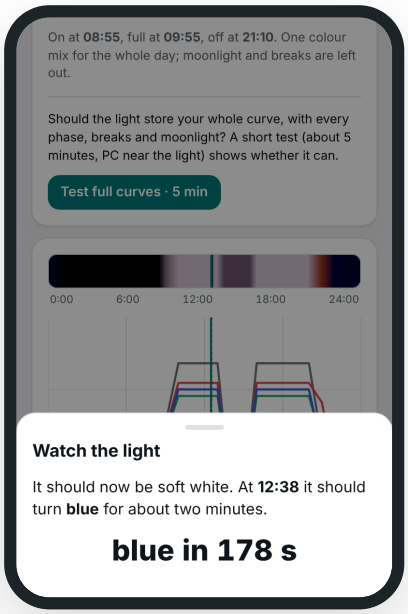
OPTIONAL

Full curves test

Can your light store your whole day program, with every phase, the midday break and moonlight? Nobody has confirmed this for the Universal WRGB yet, so Light Bridge asks the light itself. It takes about 5 minutes.



Start



Watch the light

1. **Stay near the light**
   Laptop and you next to the tank.
2. **Start the test**
   Day program → card "Stored in the light" → **Test full curves** → **Start test**.
3. **Watch**
   The light turns soft white. At the time shown it should turn **blue** for about two minutes, then white again.
4. **Answer**
   Did it turn blue? Tap **Yes**, **No** or **I missed it**.

YOUR ANSWER	WHAT HAPPENS
Yes, it turned blue	Full curves are switched on. The light runs today's program with all phases, **every day**.
No	Light Bridge tries a second way (another 5 minutes). If that does not work either, the light keeps the simple version. Nothing is lost.
I missed it	The test starts again.

Two identical lights

If both lights are the same model, you can tick **Also switch my other Chihiros to full curves**. Each light then runs its own program of today every day. If you are unsure, simply run the test on the second light too.

Afterwards your own program is always stored in the light again, whatever the result.

PEACE OF MIND

What happens if ...

SITUATION	WHAT HAPPENS
... the laptop goes to work, sleeps or is switched off?	Both lights keep running their stored programs. When the laptop is back and Light Bridge runs, it connects again by itself.
... a manual light is on when the laptop leaves?	The manual light stays on until the laptop is near that light again; then the light goes back to its program. Light Bridge shows “still in manual light”. Tip: **Back to day program** before leaving.
... the laptop leaves while a change is being stored?	The app shows “program only partly stored”. As soon as the laptop is near again, the program is stored completely.
... I walk past a light with the laptop?	Light Bridge only starts to store a program when the connection has been steady for a few seconds. Walking past does no harm.
... there is a power cut?	When the power is back, the lights start with their stored program. Their clocks may be wrong until the laptop is near them; then the clock is set automatically. Tip: after a power cut, start Light Bridge at home and walk past both lights.
... the clocks change (summer / winter time)?	The lights' clocks are corrected the next time the laptop is near them.
... the Wi-Fi is down?	The lights keep running. Only the iPhone cannot reach Light Bridge. Use the laptop directly.
... someone opens the My Chihiros app?	It may take over the light's connection. Light Bridge shows “out of range”. Close the app; Light Bridge connects again by itself.
... a neighbour has a Chihiros too?	It shows up as “new light nearby”. Nothing is changed in it unless you tap “This is mine”. Tap “Not mine” to hide it.
... the laptop is in another network, e.g. at work?	Nobody there can use Light Bridge without the code from your laptop screen. The lights at home are not affected.

Keep a backup

Gear icon → **Download backup** saves names, rooms and all day programs in one file. **Restore from file** brings them back, for example on a new laptop.

HELP

Troubleshooting

A light does not appear

- Is the light switched on? Is the laptop within a few metres?
- Is Bluetooth switched on in Windows (quick settings, bottom right)?
- Is the My Chihiros app closed on all phones?
- Gear icon → **Search now**. Did you tap “Not mine” by mistake? Then remove the light's line from the settings file, or ask whoever set up Light Bridge.

“Out of range” although the laptop is next to the light

- Close the My Chihiros app on every phone.
- Unplug the light for 10 seconds, then plug it in again.

A change shows “change waiting” for a long time

The laptop has not been near the light since the change. Take it there for a minute. The tag disappears when the program is stored.

The iPhone shows “Connect this device once”

Scan the code on the laptop screen again (overview, at the bottom, or gear icon → “Use it on your iPhone”).

The iPhone cannot open the page at all

- Is the iPhone in the home Wi-Fi (not mobile data)? Is the laptop at home and Light Bridge running?
- Windows firewall: **Windows Security → Firewall & network protection → Allow an app through firewall**. Tick “Python” for **Private**.

The light does not switch on or off at the expected time

Check the card “Stored in the light” (tab Day program): it shows the times the light really uses. The light's clock is set whenever the laptop is near; gear icon on the light → **Set the light's clock** does it at once.

Light Bridge does not start

- Was Python installed with “Add python.exe to PATH” ticked? If not, install it again with the tick.
- “Light Bridge is already running”: it is already open; the browser shows it.

FOR WHOEVER SETS IT UP

First test with your lights

Light Bridge has so far only been tested with simulated lights. Please go through this list once with the real lights and note what you see. It takes about an hour.

#	DO THIS	EXPECTED	OK?
1	Start the demo (`Start-Demo`) and click through.	Everything works with simulated lights.	
2	Start Light Bridge next to light 1. Tap **Flash**.	Light 1 blinks blue three times.	
3	**This is mine**, name and room.	“Bluetooth connected”, no “storing…” after about 15 s.	
4	Compare the tank with the card “Stored in the light”.	Brightness fits the current time.	
5	Maintenance light, then **Back to day program**.	Full light, then back to the program.	
6	Walk away with the laptop (or switch Bluetooth off).	The light keeps its program; app: “runs its own program”.	
7	Manual light 1 h, then walk away; come back after 5 min.	Away: manual light stays, app says so. Back: program again within a minute.	
8	Manual light, then close the laptop lid.	Ideally the light goes back to its program at once (sleep hook). Note what happens.	
9	Set a phase a few minutes ahead, save, watch.	The light switches/fades at the times on the card. Note whether the fade ends or starts at that time.	
10	Full curves test (page 11).	Note: blue yes/no, first or second way.	
11	Add light 2 in the other room; repeat 4–7.	Both lights work independently.	
12	iPhone: scan the code, add to Home Screen, open from the icon.	Works in the home Wi-Fi.	
13	Next morning: is each light on its program, with the right time?	Yes, without the laptop.	

Please send the results (and the file `light-bridge.log` from the settings folder, shown under gear icon → About) to whoever set up Light Bridge.

BEHIND THE SCENES

Review notes and disclaimer

Before this version was handed over, the program was checked against one question: does each light keep working in its intended mode, whatever the laptop does?

What was checked and improved

- **Laptop leaves at any moment:** tested by cutting the connection after every single command while a program is stored. The light either keeps a complete program, or the app knows and repairs it on the next visit.
- **Walking past / closing the lid:** a program is only stored on a steady connection; before sleep only the short “run your program” command is sent.
- **Honest status:** the app shows when a light was left in manual light or with an unfinished program.
- **Simple version:** moonlight no longer counts as daylight, and a light time over midnight cannot make the light shine all night.
- **Neighbour safety:** lights found nearby are only changed after “This is mine”.
- **Two lights:** each has its own connection and state; one unplugged light does not hold up the other.
- **Network safety:** other devices need the code from the laptop screen; no automatic network searches.
- **Robustness:** one Light Bridge per PC; settings saved safely; damaged files kept and reported; unsaved changes are not lost by accident.

Known limits

1. Tested with simulated lights only. The commands come from other aquarium owners' documentation.
2. Full curves are not confirmed for this light (hence the test). Without them: one colour mix per day.
3. A manual light needs the laptop nearby to end.
4. After a power cut while the laptop is away, a light's clock may be wrong until the laptop is near.
5. Windows parts (start files, sleep and close handling) are not yet tested on a real Windows laptop.
6. Bluetooth reaches a few metres; the laptop must come near each light now and then.

Disclaimer

Light Bridge is a private hobby project and comes **without any warranty** (MIT licence). It uses unofficial interfaces of Chihiros (and optionally Juwel) products that are not published by the manufacturers; a firmware update may change them. Using it changes the light program of your aquarium: please watch the tank after changes, check on your animals and plants, and keep a backup. You use it at your own risk.

Not affiliated with, endorsed or supported by Chihiros or Juwel. Product names belong to their owners.

REFERENCE

Words used in this guide

Light Bridge	The program on your laptop that stores day programs in your lights and shows one page for all of them.
Bluetooth	Short-range radio between the laptop and a Chihiros. It reaches a few metres.
Home Wi-Fi	Your wireless network at home. The laptop and the iPhone use it to talk to each other.
Day program	The light plan for one day. Each light keeps 8.
Phase	One row in a day program: a time and the colours reached at that time.
Stored in the light	What the light runs on its own: the simple version or the full curve.
Full curves	The whole day program with all phases, stored in the light (after a successful test).
Manual light	Colours set by hand. They end after the chosen time, when the laptop is near.
Maintenance light	Full light for 60 minutes, for work at the tank.
Flash	Lets a light blink blue three times, so you see which one it is.
Code (QR code)	The square pattern on the laptop screen. It connects your iPhone and contains a secret key.
Demo mode	Light Bridge with simulated lights, for trying things out.

Appendix: Juwel HeliaLux (beta)

Light Bridge can also show a Juwel HeliaLux SmartControl (Spectrum with 4 colours, LED with 2) on the same page. This part is a beta and was not the focus of this version.

1. Bring the SmartControl into your home Wi-Fi once: connect to its own Wi-Fi `HLSmartControl`, open `192.168.4.1`, Settings → Home network.
2. In Light Bridge: gear icon → **Optional: Juwel HeliaLux (beta)** → **Search this home network**, or enter its address.

The HeliaLux stores its program itself; Light Bridge reads every change back to check it. Juwel does not publish this interface, so an update could change it. There is no automatic network search, so search again at home if its address changes.